

Learning Transferable Visual Models From Natural Language Supervision

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Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at <https://github.com/OpenAI/CLIP>.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019).

Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of “text-to-text” as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image captions learn useful image representations. They converted the title, description, and hashtag metadata of images in the YFCC100M dataset (Thomee et al., 2016) into a bag-of-words multi-label classification task and showed that pre-training AlexNet (Krizhevsky et al., 2012) to predict these labels learned representations which preformed similarly to ImageNet-based pre-training on transfer tasks. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image

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	IN Top-1	IN-V2 Top-1	IN-A Top-1	IN-R Top-1	ObjectNet Top-1	IN-Sketch Top-1	IN-Vid PM0 PM10		YTBB PM0 PM10	
NS EfficientNet-L2 ^a	88.3	80.2	84.9	74.7	68.5	47.6	88.0	82.1	67.7	63.5
FixResNeXt101-32x48d V2 ^b	86.4	78.0	68.4	80.0	57.8	59.1	85.8	72.2	68.9	57.7
Linear Probe CLIP	85.4	75.9	75.3	84.2	66.2	57.4	89.1	77.2	68.7	63.1
Zero-Shot CLIP	76.2	70.1	77.2	88.9	72.3	60.2	95.3	89.2	95.2	88.5

Table 16. Detailed ImageNet robustness performance. IN is used to abbreviate for ImageNet. ^a(Xie et al., 2020) ^b(Touvron et al., 2019)

Despite this handicap, CLIP features transfer surprisingly well to this task. CLIP matches the best prior result on UCF-101 in a linear probe evaluation setting and also outperforms all other models in our evaluation suite. On Kinetics-700, CLIP also outperforms the fine-tuned I3D baseline from the original paper. Since it does not require a training stage, we report CLIP’s zero-shot performance when averaging predictions across all frames. CLIP also performs well in this setting and on Kinetics-700 its performance is within 1% of the fully supervised I3D baseline which is trained on 545000 labeled videos. Encouraged by these results, we also measure CLIP’s performance on the recently introduced RareAct dataset (Miech et al., 2020a) which was designed to measure zero-shot recognition of unusual actions like “hammering a phone” and “drilling an egg”. CLIP improves over the prior state of the art, a S3D model trained on automatically extracted captions from 100 million instructional videos, by 10 points.

While CLIP has encouragingly strong performance on the task of action recognition, we note that there are many differences between the models being compared beyond just their form of supervision such as model architecture, training data distribution, dataset size, and compute used. Further work is needed to more precisely determine what specific design decisions contribute to achieving high performance on this task.

	1km	25km	200km	750km	2500km
ISNs ^a	16.9	43.0	51.9	66.7	80.2
CPlaNet ^b	16.5	37.1	46.4	62.0	78.5
CLIP	13.9	32.9	43.0	62.0	79.3
Deep-Ret+ ^c	14.4	33.3	47.7	61.6	73.4
PlaNet ^d	8.4	24.5	37.6	53.6	71.3

Table 17. Geolocalization performance on the IM2GPS test set. Metric is percent of images localized within a given radius. Models are ordered by average performance. ^a(Muller-Budack et al., 2018) ^b(Hongseok Seo et al., 2018) ^c(Vo et al., 2017) ^d(Weyand et al., 2016)

E.4. Geolocalization

Another behavior we noticed during the development of CLIP was its ability to recognize many places and locations. To quantify this we created the Country211 dataset as described in Appendix A and report results on it throughout the paper. However it is a new benchmark so to compare with prior work on geolocalization we also report results on the IM2GPS test set from Hays & Efros (2008) in Table 17. Since IM2GPS is a regression benchmark, we guess the GPS coordinates of the nearest image in a set of reference images using CLIP’s embedding space. This is not a zero-shot result since it uses nearest-neighbor regression. Despite querying only 1 million images, which is much less than prior work, CLIP performs similarly to several task specific models. It is not, however, competitive with the current state of the art.

E.5. Robustness to Distribution Shift

Section 3.3 provides a high level summary and analysis of ImageNet-related robustness results. We briefly provide some additional numerical details in this appendix. Performance results per dataset are provided in Table 16 and compared with the current state of the art results reported in Taori et al. (2020)’s evaluation suite. Zero-shot CLIP improves the state of the art on 5 of the 7 datasets, ImageNet-R, ObjectNet, ImageNet-Sketch, ImageNet-Vid, and Youtube-BB. CLIP’s improvements are largest on ImageNet-Vid and Youtube-BB due to its flexible zero-shot capability and on ImageNet-R, which likely reflects CLIP’s pre-training distribution including significant amounts of creative content. A similar behavior has been documented for the Instagram pre-trained ResNeXt models as discussed in Taori et al. (2020).